

Alexandre Ait Ettajer

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SUMMARY

Software engineer with 4 years of experience in systems development across frontend and backend. Skilled in designing scalable architectures, leading end-to-end projects, deep understanding of software development life-cycle, and mentoring junior engineers. Focused on backend systems, distributed computing, and computational infrastructure. Passionate about solving challenging problems and continuously deepening core CS knowledge.

SKILLS

Languages: C++, JavaScript, MATLAB, Rust, Python, TypeScript, CUDA, Go, Fortran

Frameworks/Libraries: MPI, Node.js, LAPACK, PyTorch, NumPy, Matplotlib

Tools/Platforms: Docker, GitHub Actions, Git, Perforce

Collaboration: Mentorship, Cross-functional Communication, Stakeholder Alignment, Technical Presentations

EXPERIENCE

The MathWorks, Inc.

Natick, MA (Hybrid)

Software Engineer

April 2025 - Present

- Enhanced client-server messaging architecture for MATLAB visualizations, allowing internal downstream teams to easily create their own cloud-native visualizations and interactions
- Migrated backend C++ rendering primitives rulers, grid, and labels to client-side JavaScript, allowing for 25% faster interactions on local MATLAB sessions and 50% faster interactions on MATLAB Online
- Supported cross-team initiatives to allow for responsive interactions for embedded MATLAB visualizations in web browsers
- Steered working group sessions for modernizing and creating distributed visualization architectures across multiple MathWorks teams
- Designed tabular data support for core graphics (Bar, Area, Histogram), integrating SQL/Excel-style datasets
- Mentored junior engineers and communicated technical direction to senior managers across cross-functional teams
- Contributed to visual tools like `plot`, `bar`, and `area` used by millions of engineers and students worldwide

Senior Associate Software Engineer

April 2024 - April 2025

- Developed `constantplane` 3D data annotation using MATLAB and JavaScript rendering, allowing users to highlight cross-sections of volumetric data
- Designed and prototyped MATLAB's first hierarchical data viz (Sunburst Chart); led UX and design reviews
- Diagnosed and resolved backend theming and serialization bugs, ensuring compatibility across releases

Associate Software Engineer

Apr 2023 - Apr 2024

- Prototyped small-scale features, gained deep understanding of C++/XML build systems

EDG Associate

May 2022 - Apr 2023

- Solved customer issues with code generation and completed dev rotations

Sandia National Laboratory Intern

Albuquerque, NM (Remote)

Science Undergraduate Laboratory Intern

May 2021 - May 2022

- Automated visualization of parameter effects on HPC performance for MHD fluid simulations
- Presented findings to senior researchers and contributed to internal proposal pipeline

PROJECTS

Canticle for the Doomed — itch.io

Browser-based rogue-like with procedural generation, built during 7-day game jam

TypeScript, Docker, GitHub Actions

Co-developed gameplay and built hot-reload deployment pipelines

EDUCATION

Capstone Projects:

- **Distributed File System** — Implemented a distributed file system in Go using Paxos consensus for fault tolerance and data replication across multiple nodes
- **2D Magnetohydrodynamics Simulator** — Built a GPU accelerated 2D MHD simulator in C++, CUDA, and MATLAB using finite element analysis
- **3D N-Body Simulator** — Developed a parallelized 3D N-body gravitational simulator in C++ and MPI, using the Barnes-Hut algorithm to optimize performance

M.S. in Computer Science

The University of Texas at Austin

GPA: 3.56, Cum Laude

Austin, TX (December 2025)

Relevant Coursework: Parallel Systems, Advanced Operating Systems, Distributed Systems, Deep Learning, NLP

B.S. in Computer Science and B.S. in Aeronautical Engineering

Rensselaer Polytechnic Institute

GPA: 3.93, Summa Cum Laude

Troy, NY (May 2022) Relevant Coursework: Data

Structures, Algorithms, Computer Architecture, Operating Systems, Numerical Methods, Finite Element Analysis, Numerical Linear Algebra, Numerical Solutions to PDEs